innovate achieve lead

BITS Pilani
Pilani Campus

SURVEYING (CE F213)
(Tacheometric Surveying)

Tacheometric Surveying

Tacheometric surveying is defined as an optical distance measurement method in which both horizontal distance and vertical distance can be determined by taking observations with an instrument known as a **Tachometer**.

Tacheometric surveying is also known as Tacheometry, Telemetry and Tachymetry.

To learn about

(i) What is Tacheometry
(ii) Instruments used
(iii) Anallactic Lens
(iv) Determination of horizontal and vertical distances
(v) Fixed hair method
(vi) Movable hair method
(vii) Tangential method

Advantages

1. Chaining is eliminated completely
2. It can be used where obstacles are present
3. Can be adopted where direct levelling and chaining are not possible
4. Very rapid(though not very accurate)
5. Contour maps can be prepared very easily

Purposes of Tachometry

- Prepare contour map.
- Used in hydrographic survey.
- Location survey for road, railway, reservoir etc.
- Checking of the distance which measured with the help of the tap.
- To measure the horizontal distance at which the distance measured by the tap or chain is difficult.

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The principle of tacheometry is based on the property of isosceles triangle.

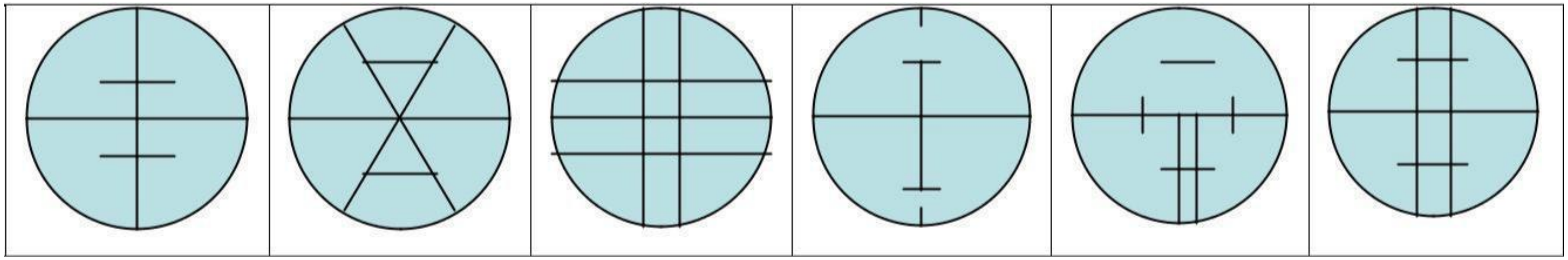
Instruments used

1. Tacheometer- (It can be a transit theodolite, the diaphragm of which is furnished with stadia wires(two horizontal wires) in addition to cross wires.)

2. Levelling Staff- (If the distance is less than 100m, it can be used, the least count is 5mm i.e. 0.005m

3. Stadia rod- (Kind of levelling staff with clear and bold graduation with least count of 10mm or 0.01m)

Different types of diaphragm



Methods of Tacheometry

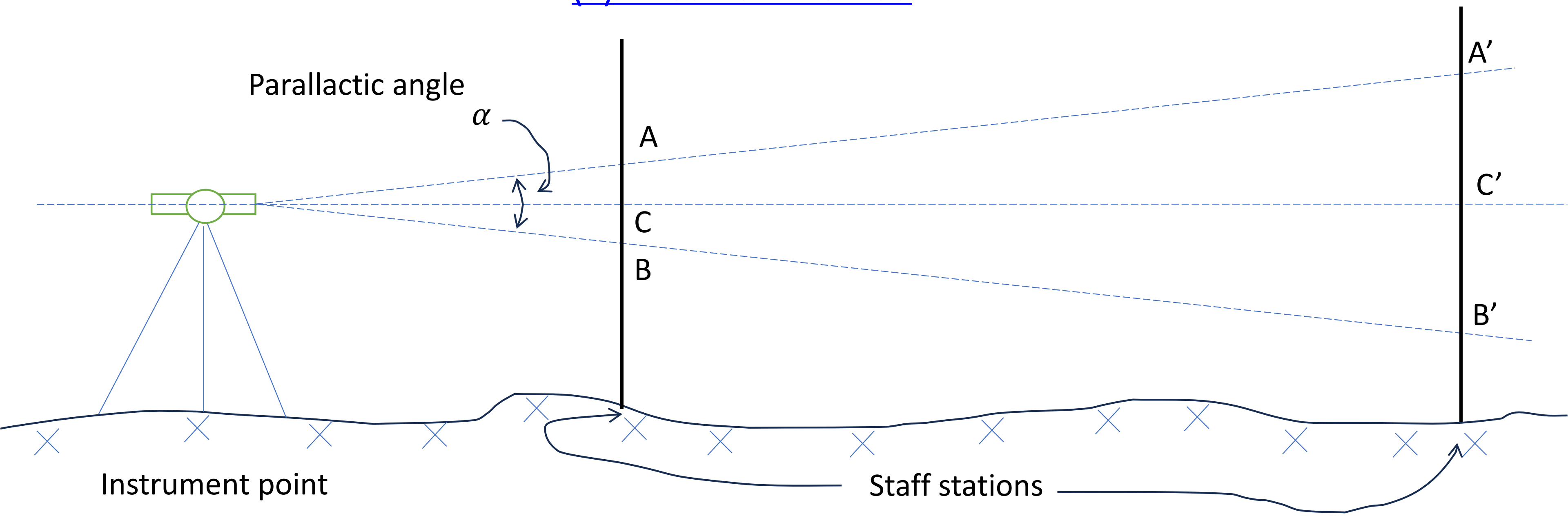
(i) Stadia Method
(ii) Tangential Method
(iii) Range Finding method

Stadia Method

(a) Fixed hair method
(b) Movable hair method(Subtense method)

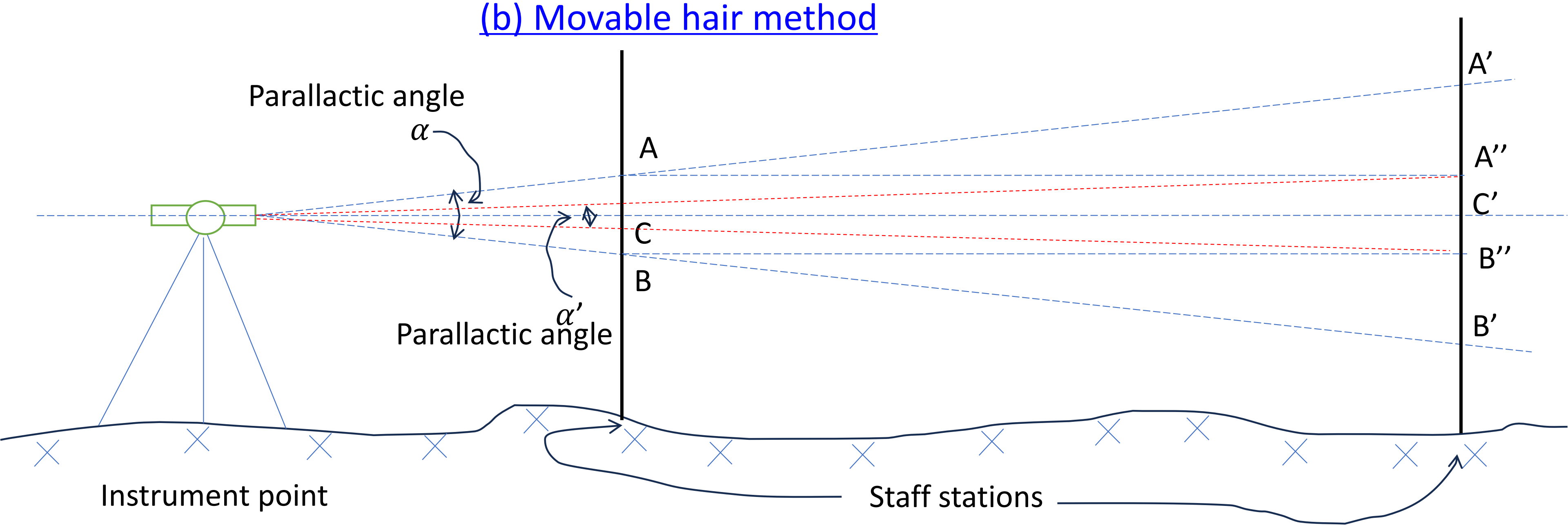
Stadia Method

(a) Fixed hair method



Parallactic angle(α) –defined by the stadia wires is fixed and the staff intercept(AB or A’B’) is varied.

(b) Movable hair method



Parallactic angles(α or α') –defined by the stadia wires are varied and the staff intercepts(AB and A''B'') are same. i.e. AB=A''B''

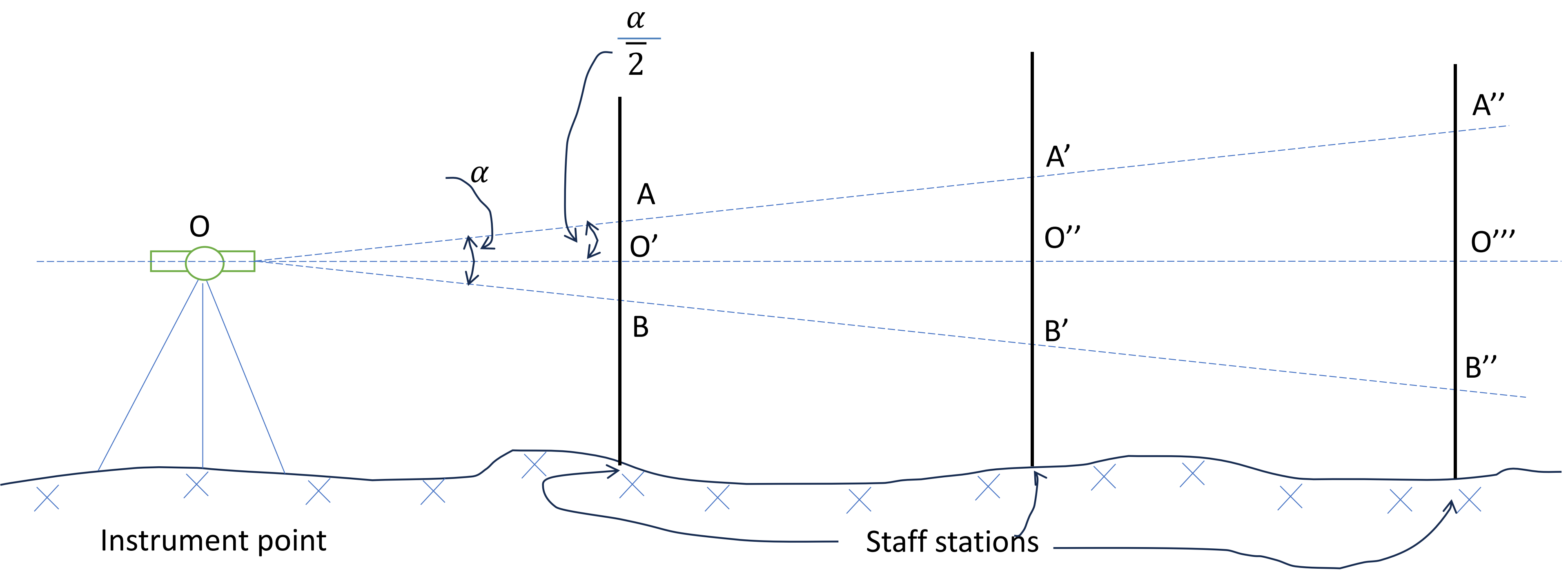
Tangential Method

Vertical angles and the staff readings are measured by the cross-wires only. Stadia wires are not used at all. This method is similar to trigonometric levelling. Height and Distance calculations using trigonometry.

Range Finding method

To find the horizontal distance and direction of a line without going to the far end of the line.

Stadia Method (Fixed hair method)



$$\frac{OO'}{AB} = \frac{OO''}{A'B'} = \frac{OO'''}{A''B''} = \frac{1}{2}\,Cot\left(\frac{\alpha}{2}\right) = K\ (constant)$$

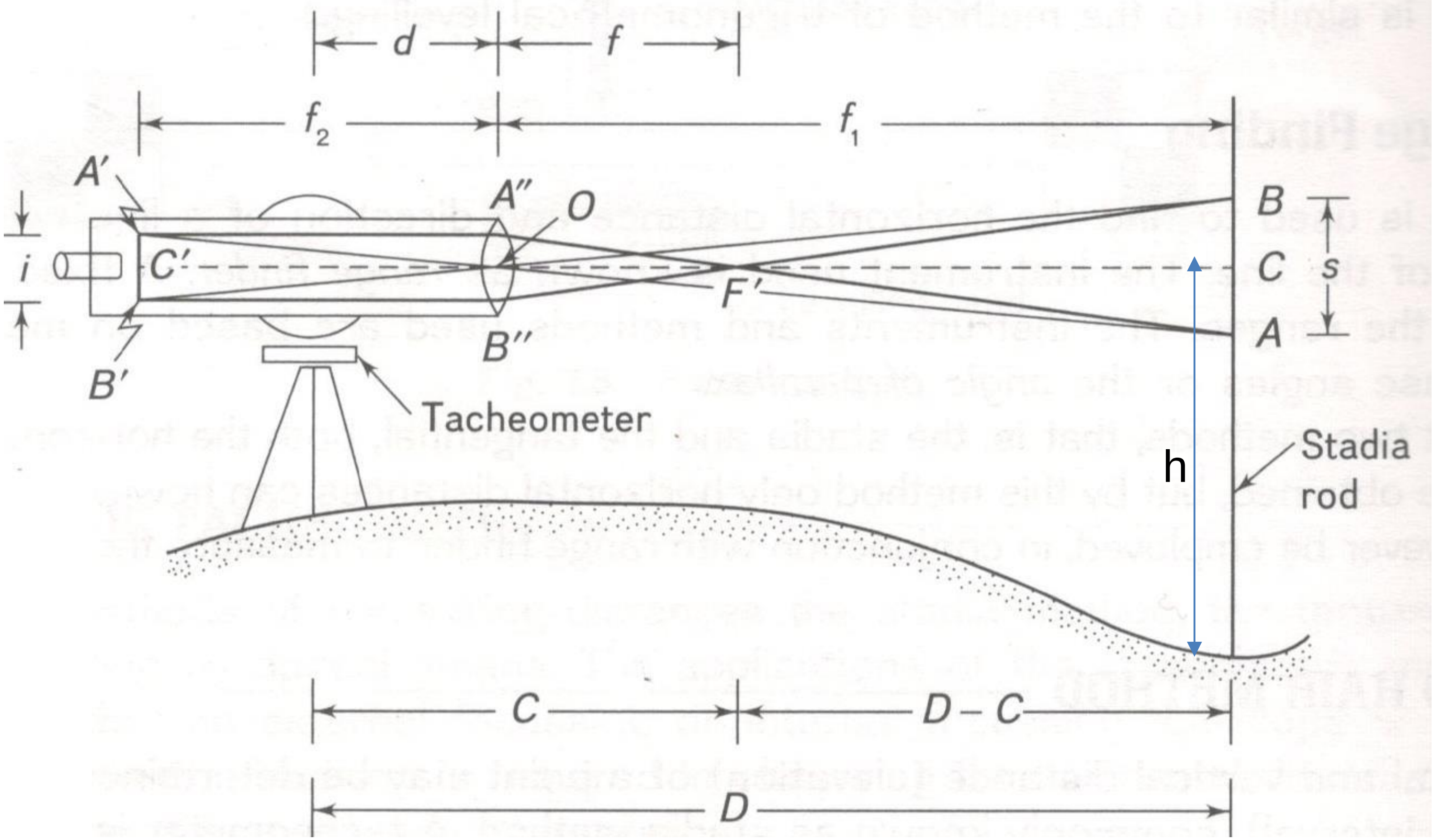
If the parallactic angle is known, the horizontal distance can be calculated.

Q1. The parallactic angle is given as 1 degree and stadia intercept is observed as 20cm in a tacheometry surveying. Find the horizontal distance between the instrument station and the stadia station.

Solution: Ans is 11.457m

If the parallactic angle is not known**(and line of Sight is horizontal)**

(Principle of stadia method)



Where,

O- Optical center of object glass

A', B' – Stadia wires

A'B' = i = **stadia interval**

C'- Horizontal cross-wire position

s- Stadia intercept (Or Staff intercept)

f- focal length of object glass

f_1- conjugate focal length of Lens(Optical centre to staff) – Object dist.

f_2- conjugate focal length of Lens(Optical centre to staff image) – Image dist.

d- vertical axis of Tacheometer to optical centre

D- Vertical axis to stadia distance

From triangle OAB and triangle OA’B’ (both are similar Triangle)

$$\frac{AB}{A'B'} = \frac{OC}{OC'}$$

$$\frac{s}{i} = \frac{f_1}{f_2}$$

Using Lens formula (externally focusing type telescope)

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \Rightarrow \frac{f_1}{f} = \frac{f_1}{f_1} + \frac{f_1}{f_2} \Rightarrow \frac{f_1}{f} = 1 + \frac{f_1}{f_2} \Rightarrow \frac{f_1}{f} - 1 = \frac{f_1}{f_2}$$

So $\frac{s}{i} = \frac{f_1}{f} - 1 \Rightarrow \frac{s}{i} + 1 = \frac{f_1}{f} \quad \Rightarrow \quad \frac{fs}{i} + f = f_1$

$$\frac{fs}{i} + f + d = f_1 + d$$

$$\Rightarrow \quad s\left(\frac{f}{i}\right) + (f + d) = f_1 + d$$

$$\Rightarrow \quad Ks + C = D$$

$\Rightarrow \quad D = Ks + C$, It is horizontal distance

K- Multiplying constant
C- Additive constant

RL of Staff station = RL of instrument axis – Horizontal cross-wire reading(h)

K is 100 or multiple of 100 normally (Stadia wires are accordingly spaced)

Anallactic Lens

If an anallactic lens is built into the telescope and proper adjustment are made to make all observations to the centre of the instrument axis, then C becomes 0(zero).

$C=0$

$D=Ks$

Q2. Find the stadia constants form the following observations

Instrument Station	Staff reading on Stations	Distance(m)	Stadia Reading	
			Lower	Upper
O	A	150	1.255	2.750
	B	200	1.010	3.005
	C	250	0.755	3.265

Solution

Line of Sight is inclined (i.e. Line of sight not horizontal)

Line of sight can be inclined up

Line of sight can be inclined down

Staff Vertical

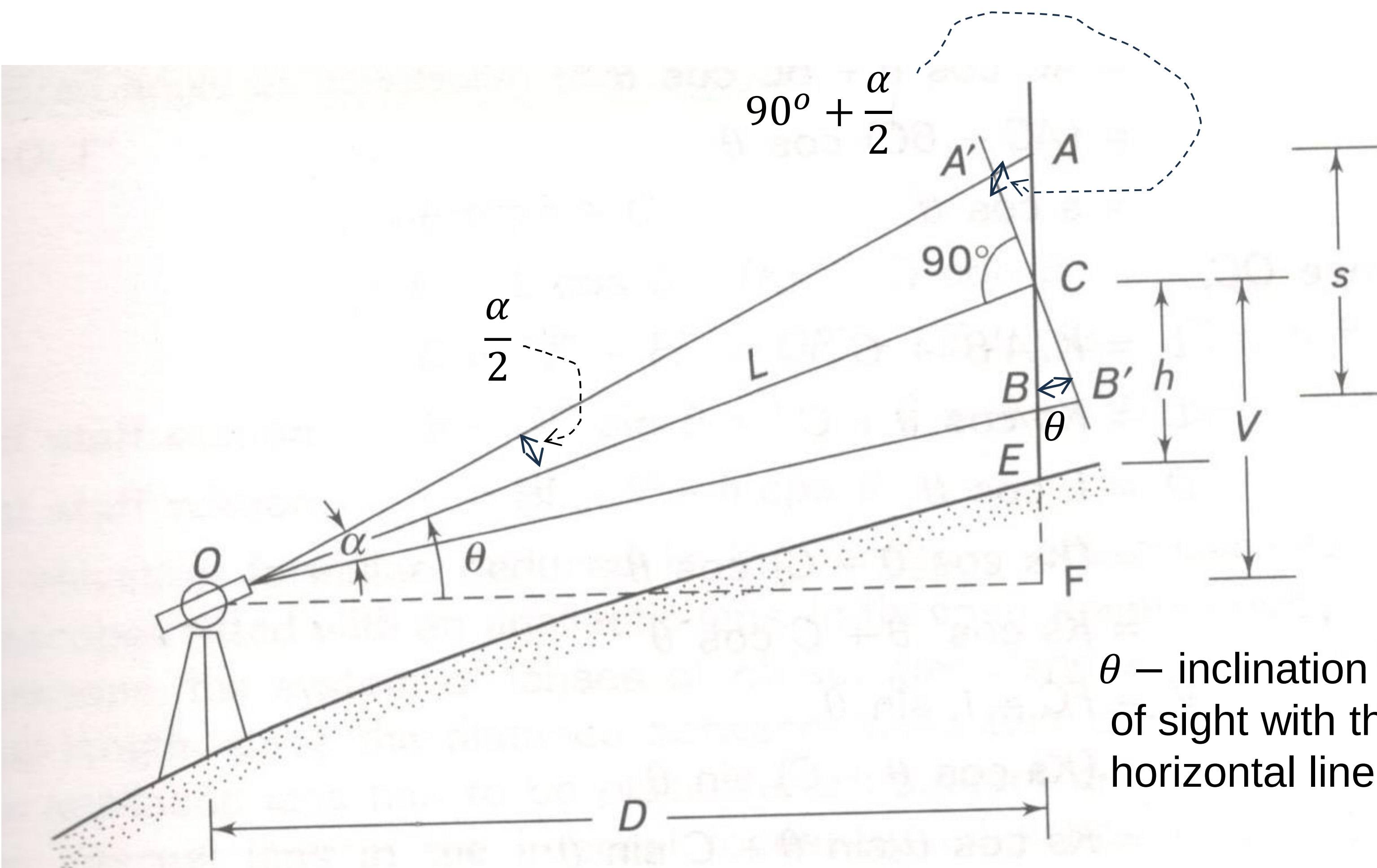
Staff perpendicular to line of Sight

Staff Vertical

Staff perpendicular to line of Sight

Line of sight can be inclined up

Staff Vertical



θ – inclination of the line of sight with the horizontal line

A'B' is an imaginary perpendicular line to the inclined line of sight,

So, The inclined length L can be found out as

L= K(A'B') + C , as per the previous formula,

However, A'B' is not known,

In the triangle AA'C the angle AA'C is $(90^o + \frac{\alpha}{2})$

Actually, the parallactic angle (α) is very small in comparison to 90^o , (α) is around 17 minutes.

So, we can assume the triangle AA'C to be an right angle triangle,

$$\cos\theta = \frac{A'C}{AC} \Longrightarrow A'C = AC\ \cos\theta$$

Similarly, from triangle BB'C, we get

$$B'C = BC\ \cos\theta$$

$$A'B' = A'C + B'C = AC \cos\theta + BC \cos\theta = (AC + BC)\cos\theta = s \cos\theta$$

$$So, L = K(A'B') + C = Ks\cos\theta + C$$

$$Now, the\ horizontal\ distance, D = L\cos\theta = (Ks\cos\theta + C)\ \cos\theta$$

$$D = L\cos\theta = Ks\ cos^2\theta + C\ cos\theta$$

$$Now, the\ vertical\ distance, V = L\sin\theta = (Ks\cos\theta + C)\ \sin\theta$$

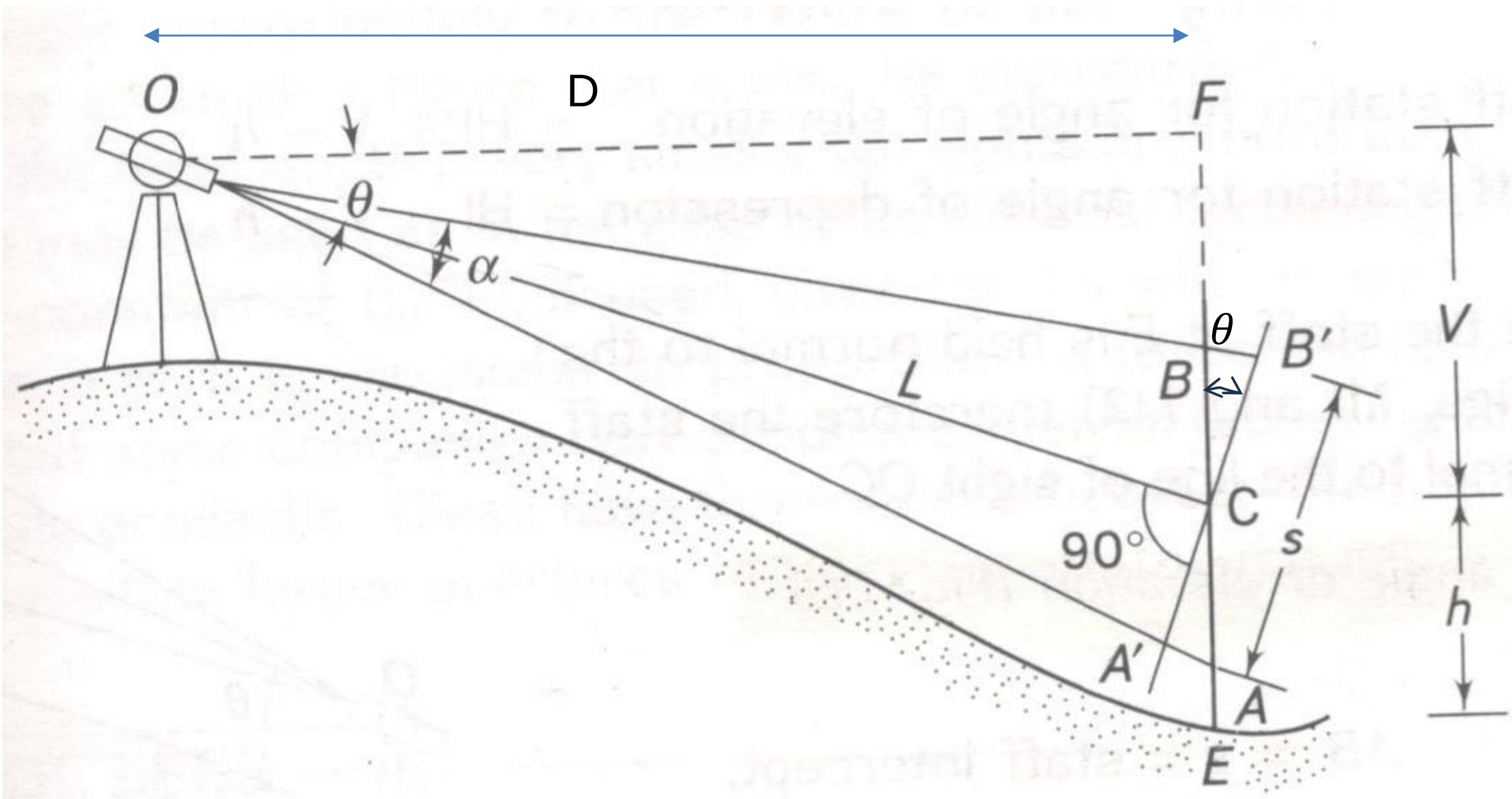
$$V = Ks\cos\theta\sin\theta + C\sin\theta$$

$$V = \frac{1}{2}Ks\sin 2\theta + C\sin\theta$$

RL of Staff station = Height of the instrument + V - h

Line of sight can be inclined down

Staff Vertical



In this case also,

$$the\ horizontal\ distance, D = L\cos\theta = (Ks\cos\theta + C)\ \cos\theta$$

$$D = L\cos\theta = Ks\ cos^2\theta + C\ cos\theta$$

$$the\ vertical\ distance, V = L\sin\theta = (Ks\cos\theta + C)\ \sin\theta$$

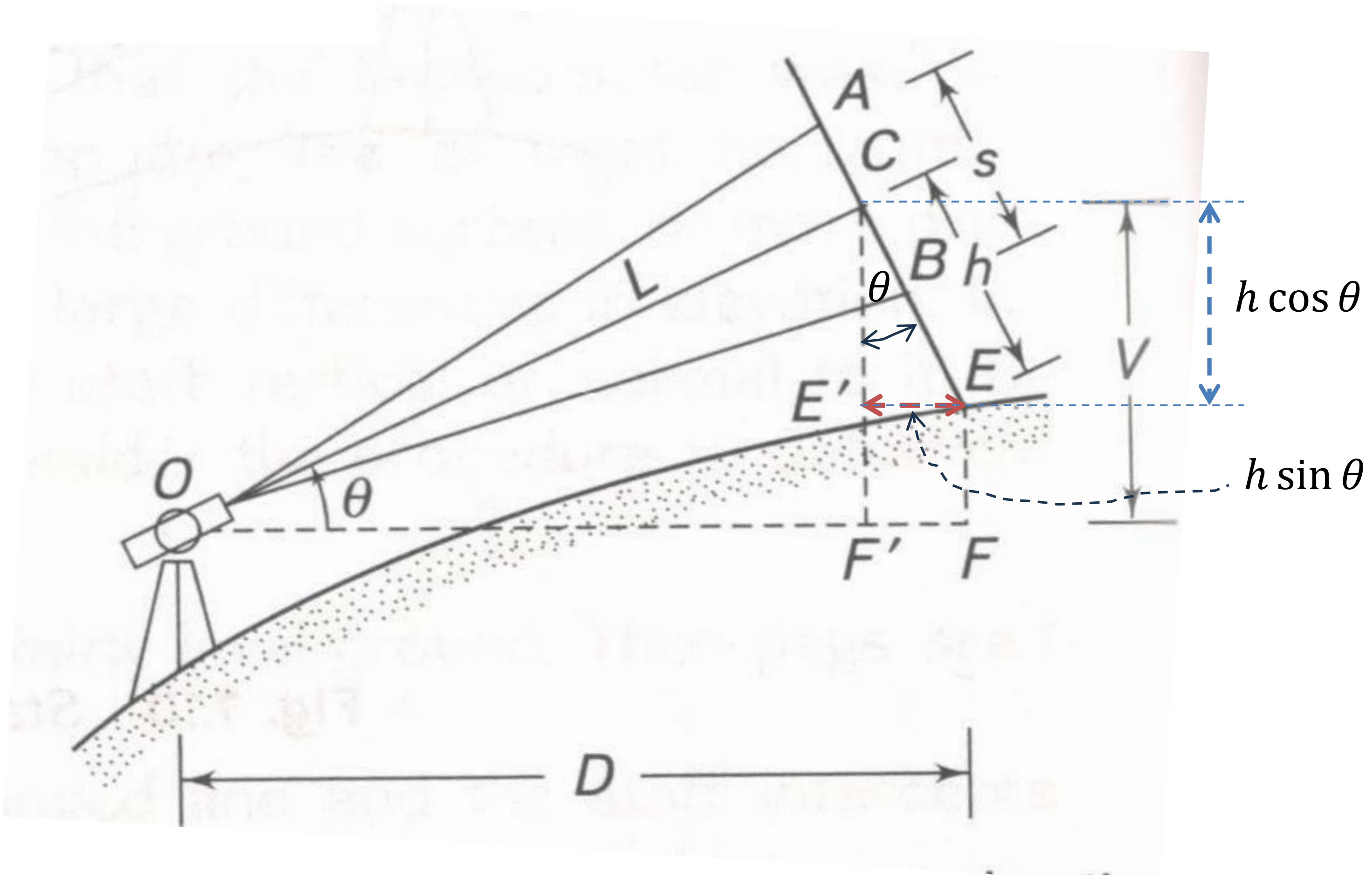
$$V\ = Ks\cos\theta\sin\theta + C\sin\theta$$

$$V\ = \frac{1}{2}Ks\sin 2\theta + C\sin\theta$$

RL of Staff station = Height of instrument - V - h

Line of sight can be inclined up

Staff perpendicular to line of Sight



In this case,

$$the\ horizontal\ distance, D = OF' + F'F = L\cos\theta + h\ \sin\theta$$

$$here,\ L = Ks + C$$

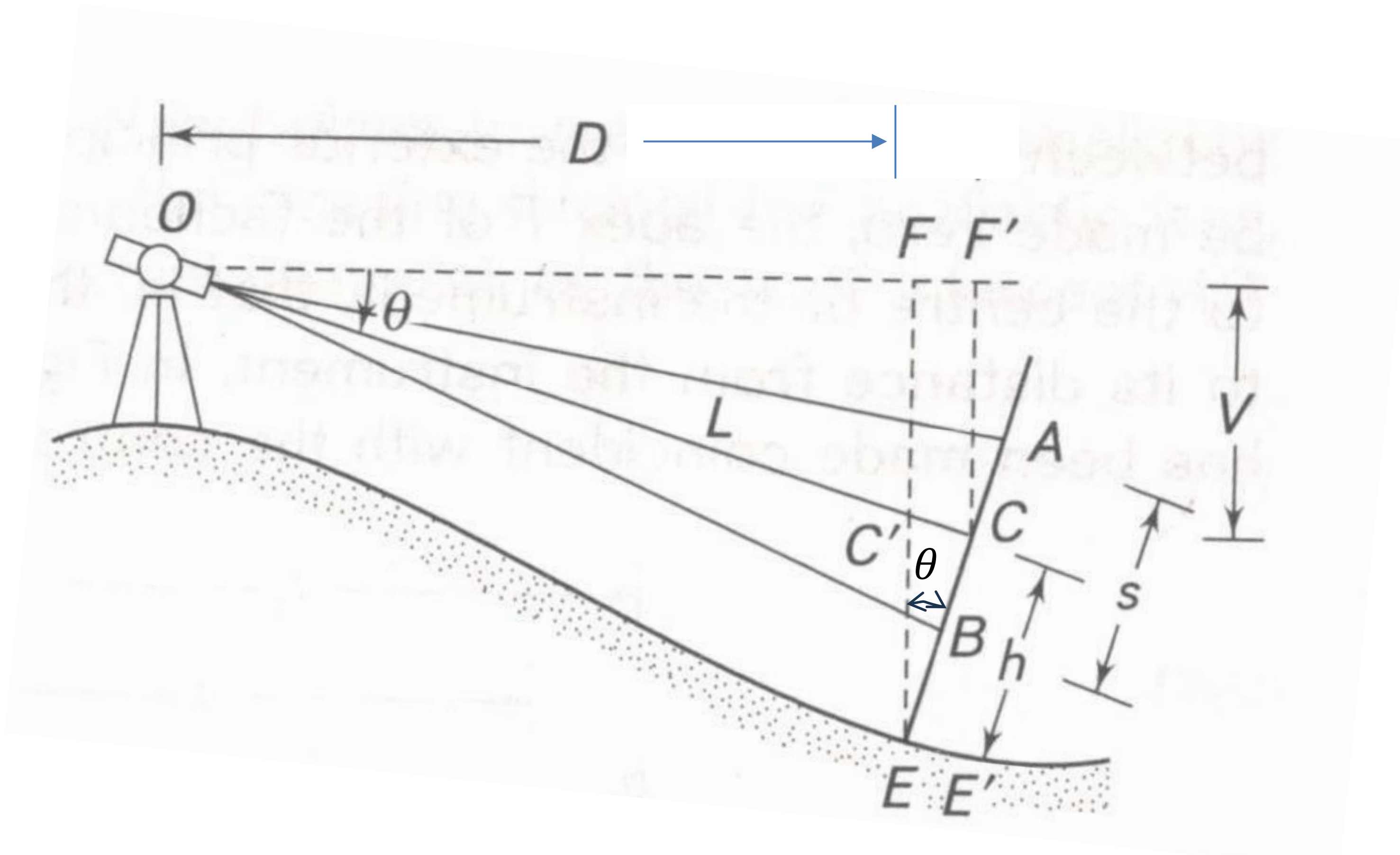
$$So, D = (Ks + C)\cos\theta + h\ \sin\theta$$

$$the\ vertical\ distance, V = L\sin\theta = (Ks + C)\ \sin\theta$$

RL of Staff station = RL of instrument axis + V – h cos θ

Line of sight can be inclined down

Staff perpendicular to line of Sight



In this case,

$$the\ horizontal\ distance, D = OF' - F'F = L\cos\theta - h\ \sin\theta$$

$$here,\ L = Ks + C$$

$$So, D = (Ks + C)\cos\theta - h\ \sin\theta$$

$$the\ vertical\ distance, V = L\sin\theta = (Ks + C)\ \sin\theta$$

RL of Staff station = RL of instrument axis - V – h cos θ

Q3.To determine the constants of a tacheometer the follow observations were made. The line of sight is horizontal and the staff is vertical. Determine the Constants. Find the RLs of Point B, C and D.

Instrument Station	Height of the Instrument (m)	Staff Station	Horizontal Distance(m)	Staff Reading(m)	Remark
A	1.35	B	250.4	1.085,2.335,3.585	RL of the point A is 200m
		C	124.25	0.500,1.125,1.750	
		D	205.02	0.755,1.760,2.765	

Sol:

We have

$$D = Ks + C$$

Equation (1) $250.4 = K \times 2.5 + C$

Equation (2) $124.25 = K \times 1.25 + C$

Equation (3) $205.02 = K \times 2.01 + C$

Taking (1) and (2)

$$K = 100.92, \qquad C = -1.9$$

Taking (2) and (3)

$$K = 106.276, \qquad C = -8.595$$

Taking (1) and (3)

$$K = 92.612, \qquad C = 18.869$$

Average of 3 constants

$$K = 99.936, \ C = 2.791$$ [3.5 +3.5]

Height of collimation line = 200+1.35 = 201.35m

RL of B = 201.35 – 2.335 = 199.015 m [1]

RL of C = 201.35 – 1.125 = 200.225 m [1]

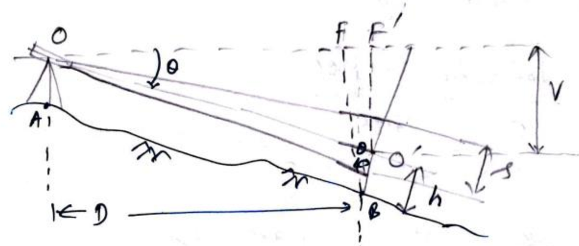
RL of D = 201.35 – 1.760 = 199.590 m [1]

Q4:A tacheometer was kept at A and the staff was kept at two different points B and C. The staff was kept normal to the line of sight during the observations. The data collected is mentioned below,

Instrument at	Staff at	Vertical angle	Staff reading in m
A	B	$-5^{o}\,0'\,37''$	1.235, 1.785, 2.335
	C	$3^{o}\,12'\,40''$	0.875, 1.475, 2.075

The RL of the instrument axis is 300.135m. The WCB of line AB is $23^{o}\,22'$ and of line AC is $99^{o}\,28'\,50''$. Find the distances AB, AC and BC. Also find the gradient of line BC. Take K=100 and C=0.2.

Q.3 1st observation.



Horizontal distance $AB = D_1 = OF' - OF = OO'\cos\theta - h\sin\theta$

$= (ks + C)\cos\theta - h\sin\theta$

$= [(100 \times 1.1) + 0.2]\cos(5^\circ 0' 37'') - 1.785\sin(5^\circ 0' 37'')$

$= \cancel{108.219\,m}$ $109.623\,m$ (2)

$V_1 = OO'\sin\theta = (ks + C)\sin\theta = 110.2\sin(5^\circ 0' 37'')$

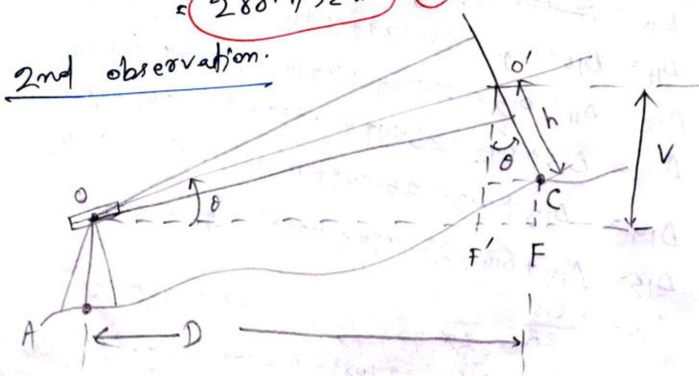
$= 9.624\,m$ (2)

R.L. of B = elevation of Ins. axis − V − h cos θ

$= 300.135 - 9.624 - 1.785\cos(5^\circ 0' 37'')$

$= 288.732\,m$ (2)

2nd observation.



(5)

$AC = OF' + FF' = OO'\cos\theta + h\sin\theta$

$= (ks + C)\cos\theta + h\sin\theta = (100 \times 1.2 + 0.2)\cos(3^\circ 12' 40'') + 1.475 \sin(3^\circ 12' 40'')$

$= 120.093\ m$ (2)

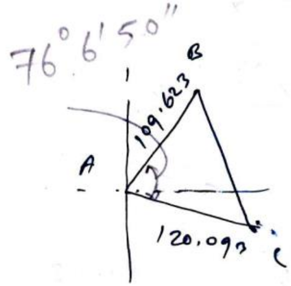
$V = OO'\sin\theta = (ks + C)\sin\theta = 120.2 \sin(3^\circ 12' 40'')$

$= 6.733\ m$ (2)

R.L. of C = R.L. of inst. axis $+ V - h\cos\theta$

$= 300.135 + 6.733 - 1.475\cos(3^\circ 12' 40'')$

$= 305.395\ m$ (2)



$BC = \sqrt{109.623^2 + 120.093^2 - 2 \times 109.623 \times 120.093 \times \cos(76^\circ 6' 50'')}$

$= 141.845\ m$ (2)

(1)

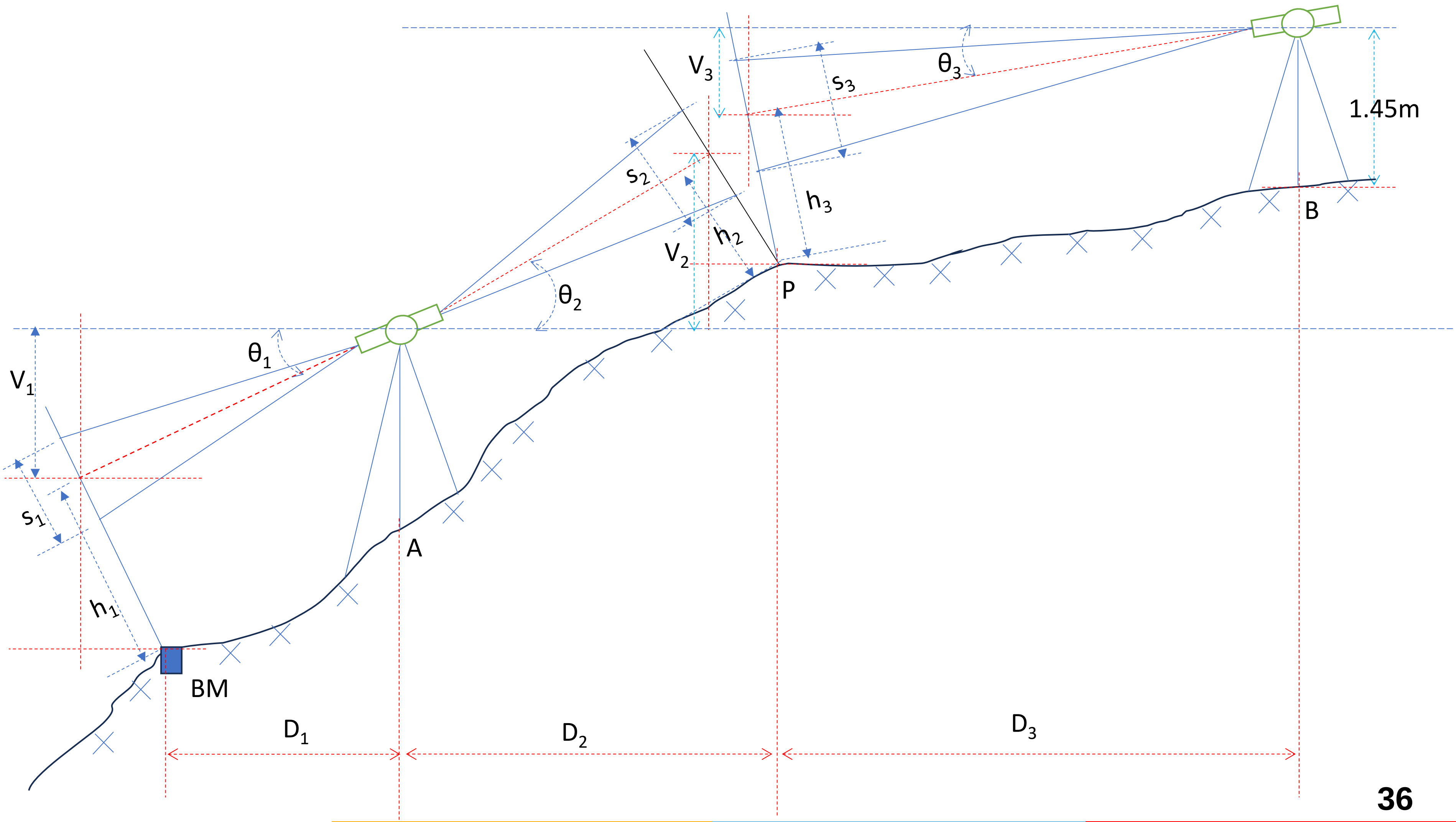
Gradient of BC $= \dfrac{305.395 - 288.732}{141.845} = 0.117 = \dfrac{1}{8.512}$

C is at higher level than B

Q4:The following observations are taken with a tacheometer fitted with an *anallactic lens* keeping the staff normal to the line of sight. The constant of the tacheometer is 100.

Find the horizontal distance between A and BM, A and P, B and P, RL of B.

Instrument station	Height of the Instrument(m) from ground level	Staff Station	Vertical angles	Staff readings(m)	Remark
A	1.25	BM	$-5^{o}\ 20'$	1.005, 2.005, 3.005	RL of BM is 500m
A	1.25	P	$7^{o}\ 40'$	0.075, 0.825, 1.575	
B	1.45	P	$-4^{o}\ 50'$	1.235, 1.885, 2.535	CP


V3
θ3
s3
1.45m
s2
h3
B
h2
V2
θ2
P
θ1
V1
s1
A
h1
BM
D1
D2
D3

V- vertical distance from line of collimation to staff middle hair reading
h- distance from foot of the staff to middle hair reading along the staff

BM

1st observation. (Instrument at A staff at Bm) (Angle of Depression)

$s_1 = 2.000\,m$

$\theta_1 = 5°20'$

$= (5.333)°$

$h = 2.005\,m$

$D_1 = \quad = L\cos\theta_1 - h_1\sin\theta_1$

$D_1 = Ks\cos\theta_1 - h_1\sin\theta_1$

$= 100 * 2 * \cos(5.333) - 2.005\sin(5.333)°$

$= 200 * 0.995 - 0.186$

$D_1 = 198.814\,m$

$V_1 = L\sin\theta_1 = Ks\sin\theta_1 = 100 * 2 * \sin(5.333)°$

$= 18.588\,m$

R.L. of Line of Collimation = B.M. + $V_1 + h_1\cos\theta_1$

$= 500 + 18.588 + 2.005 \cos(5.333)$ ②

$= 520.584\,m$

2nd observation. (Angle of elevation)

$\theta_2 = 7^\circ 40' = (7.666)^\circ$, $(s_2) = 1.5^m$ / $h_2 = 0.825\,m$

$D_2 = L\cos\theta_2 + h_2 \sin\theta_2 =$

$= Ks\cos\theta_2 + h_2 \sin\theta_2$

$= 100 * 1.5 * \cos(7.666) + 0.825 \sin(7.666)$

$= 148.769\,m$

$V_2 = L\sin\theta_2 = (Ks)\sin\theta_2 = 100 * 1.5 \sin(7.666)$

$= 20.009\,m$

R.L. of Point P = R.L. of line of collimation $+ V_2 - h_2\cos\theta_2$

$= 520.584 + 20.009 - 0.825\cos(7.666^\circ)$

$= 539.775\,m.$

3rd observation (Angle of depression)

$s_3 = 1.3\,m$

$\theta_3 = 4^\circ 50' = 4.833^\circ$

$h_3 = 1.885\,m$

$D_3 = L_3\cos\theta - h_3 \sin\theta_3$

$= KS\cos(4.833) - 1.885\sin(4.833)$

$= 100 * 1.3 * \cos(4.833) - 1.885 \sin(4.833)$

$= 129.537 - 0.158 = 129.379\,m$

$V_3 = L\sin\theta_3 = 100 * 1.3 * \sin(4.833) = 10.952\,m$

R.L. of line of collimation = R.L. of point P $+ V_3 + h_3\cos\theta_3$

$= 539.775 + 10.952 + 1.885\cos(4.833)$

$= 552.605\,m$

R.L. of Point B = 552.605 − 1.45

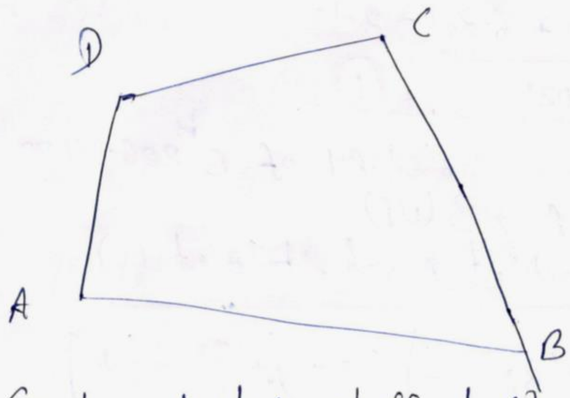
$= 551.155\,m$

Q5. In a four-sided land patch, a tacheometer was kept at one of the vertices and readings were taken on staff held at other three consecutive points. The data collected was as below:

Instrument at	Staff at	Vertical angle	Staff readings in m
A	B	2^{o} 35’	1.350, 2.100,2.850
	C	-5^{o} 39’ 38”	2.135,2.385,2.635
	D	0^{o} 57’ 35”	1.875,2.375,2.875

The staff was held vertical during the measurement. Take K=100 and C=0.4. The RL of the instrument station is 200m and the height of the instrument is 1.35m. Check whether the land patch is rectangular. Find the gradients of the line AB, AC and AD.

No-1



4-Sided land Patch.

$K = 100$, $C = 0.4$ } Constants

1st (Instrument at A, staff at B)

$\theta = 2^\circ 35' = \left(2 + \frac{35}{60}\right)^\circ = 2.583^\circ$ (angle of elevation)

s = staff/stadia intercept $= 2.850 - 1.350 = 1.500\,m$

$D_1 = AB$ (horizontal distance between A & B)

$= (K s \cos\theta + C)\cos\theta = K s \cos^2\theta + C\cos\theta$

$= 100 * 1.5 \cos^2 2.583 + 0.4 \times \cos 2.583 = 149.695 + 0.399$

$= 150.094\,m$

$\boxed{AB = 150.094\,m}$ (1)

V_1 = vertical distance ~~between A and B~~ Line of Collimation and ~~staff reading~~ central hair

$= (K.s.\cos\theta + C)\sin\theta = \frac{1}{2} K s \sin 2\theta + C \sin\theta$

$= \frac{1}{2} * 100 * 1.5 * \sin(2*2.583) + 0.4 * \sin 2.583^\circ$

$\boxed{V_1 = 6.753 + 0.018 = 6.771\,m}$ (1)

R.L. of A is 200m, height of instrument is 1.35m

R.L. of Collimation line = 200 + 1.35 = 201.35m

So R.L. of B = R.L of Collimation line + $V_1 - h_1$

(h_1 = central h... reading)

$= 201.350 + 6.771 - 2.100$

$\boxed{R.L. \text{ of } B = 206.021 \text{ m}}$ (1)

R.L. of A is 200 m and R.L. of B is 206.021 m

A → B (UP)

2nd (instrument at A and staff at C.)

$\theta = -5° 39' 38'' = -\left[(5° 39') + \frac{38}{60} \text{ minutes}\right]$

$= -(5° 39' + (0.633)')$

$= -(5° 39.633') =$

$= -(5.66)°$ — -ve sign indicates angle of depression.

$s = 2.635 - 2.135 = 0.500$ m

here, $D_2 = AC = Ks\cos^2\theta + C\cos\theta$

$= 100 * 0.5 * \cos^2\theta + 0.4\cos\theta$

$= 49.756 + 0.398$

So, $\boxed{D_2 = 50.154 \text{ m}}$ (1)

$V_2 = \frac{1}{2} Ks \sin 2\theta + C\sin\theta$

$= \frac{1}{2} * 100 * 0.5 * \sin 2\theta + 0.4\sin\theta$

$\boxed{V_2 = 4.946 \text{ m}}$ (1)

R.L. of C = R.L. of Collimation line $- V_2 - h_2$

$= 201.350 - 4.946 - 2.385$

$\boxed{R.L. \text{ of } C = 194.019 \text{ m}}$ (1) A → C (down)

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3rd (Instrument at A, staff at D)

$s = 2.875 - 1.875 = 1.000\,m$

$\theta = 57' 35'' = (0.959)^\circ$

$D_3 = K s \cos^2\theta + C \cos\theta = AD$

$= 100 * 1 * \cos^2(0.959) + 0.4 \cos(0.959)$

$\boxed{D_3 = 99.971 + 0.399 = 100.37\ m}$ (1)

$V_3 = \frac{1}{2} K s \sin 2\theta + C \sin\theta$

$= \frac{1}{2} * 100 * 1 * \sin 2\theta + 0.4 \sin\theta$

$\boxed{V_3 = 1.673 + 0.006 = 1.679\,m}$ (1)

R.L. of point D = R.L. of line of collimation $+ V_3 - h_3$

$= 201.350 + 1.679 - 2.375$

$\boxed{\text{R.L. of Point D} = 200.654\ m}$ (1)

A → D (up)

To check whether rectangular patch?

For ABCD to be rectangular [Considering horizontal Distances]

AD should be equal to BC, Let's take (consider)

AD = BC = 100.37 m. If this then

$AB^2 + BC^2$ should be equal to AC^2

(32602.34 m2) Not coming equal (9515.42 m2)

2

Also AC should be greater than AB and BC individually. But AC is less than AB and less than BC. So it is not a rectangular patch.

Gradient of AB

$$\frac{206.021-200}{150.094} = \frac{6.021}{150.094} = \frac{1}{24.92}$$

$$= 0.0401$$

A to B is up.

Gradient of AC.

$$\frac{200-194.019}{50.154} = \frac{5.981}{50.154} = \frac{1}{8.385}$$

$$= 0.1192$$

A to C is down

Gradient of AD

$$\frac{200.654-200}{100.370} = \frac{0.654}{100.370} = \frac{1}{153.47}$$

$$= 0.00651$$

A to D is up

4

Stadia Method (Movable hair method)

Stadia **interval** is variable.

Stadia/Staff **intercept** is fixed.

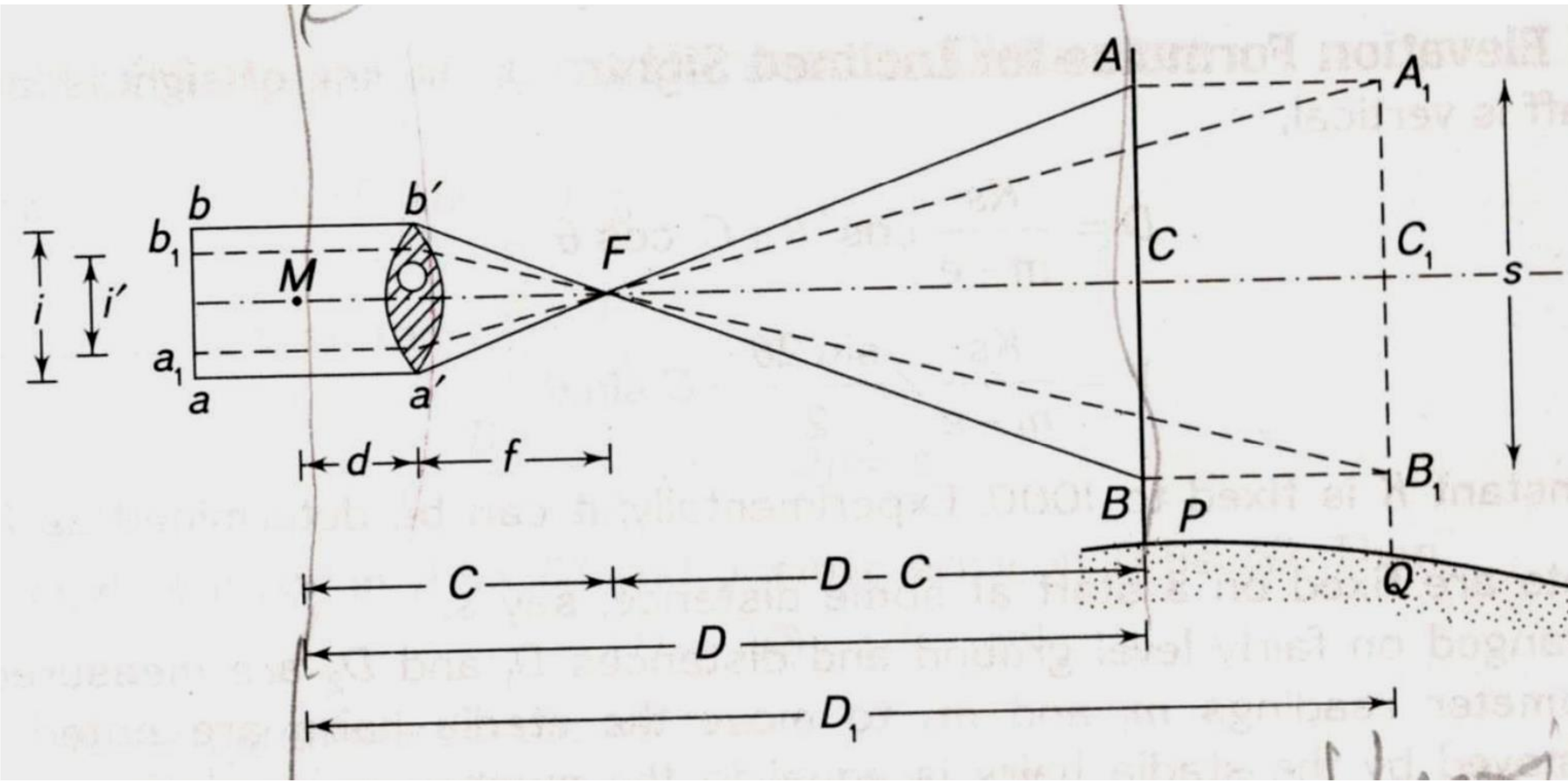
The Stadia/Staff **intercept** is called base.

Two methods are there

(i) Vertical base subtense method

(ii) Horizontal base subtense method

Vertical base subtense method (for horizontal line of sight)



Triangle ABF and Triangle a'b'F (Similar triangles)

$$\frac{FC}{AB} = \frac{OF}{a'b'} \Rightarrow \frac{FC}{s} = \frac{f}{i} \Rightarrow FC = \frac{fs}{i}$$

$$D = FC + FM = \frac{fs}{i} + (f + d)$$

Micrometer is used to find the value of i (stadia interval), i=mp

m=no of revaluations of micrometer

P=pitch of the micrometer screw

$$D = \frac{fs}{mp} + (f + d)$$

$$\boxed{D = \left(\frac{f}{p}\right)\frac{s}{m} + (f + d) \Rightarrow D = \frac{Ks}{m} + C}$$

$$\boxed{K = \frac{f}{p} \text{ and } C = f + d}$$

K- Multiplying constant

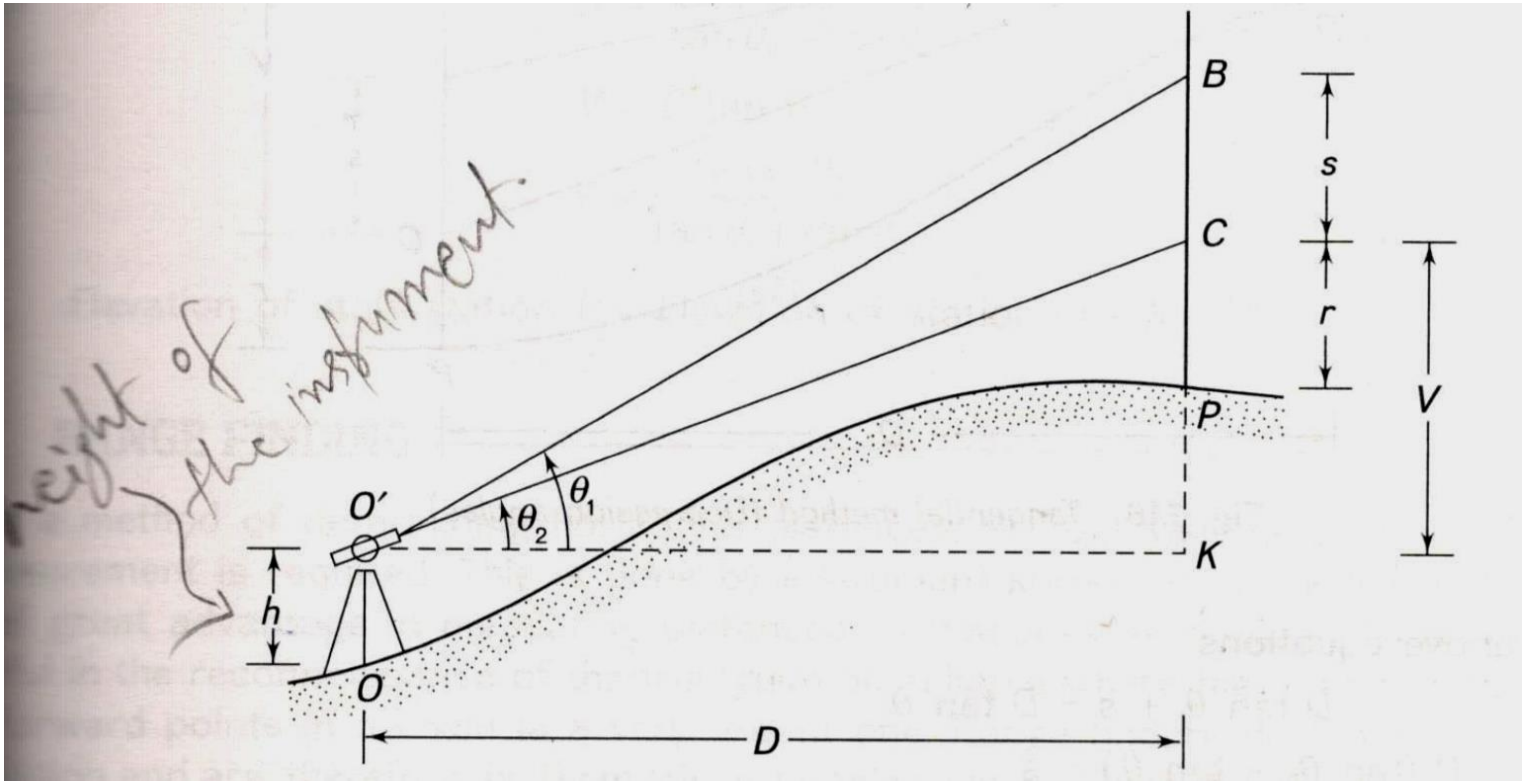
C- Additive constant

If e= index error is present, then

$$\boxed{D = \frac{Ks}{m - e} + C}$$

Tangential Method (Three Cases) Only cross-wire reading are necessary, No stadia wires necessary.

Case-I (Both angles are angle of elevation)



$$\tan\theta_2 = \frac{V}{D} \Rightarrow V = D\ \tan\theta_2$$

$$\tan\theta_1 = \frac{V+s}{D} \Rightarrow V + s = D\ \tan\theta_1$$

$$D\tan\theta_2 + s = D\ \tan\theta_1$$

$$D\tan\theta_1 - D\ \tan\theta_2 = s$$

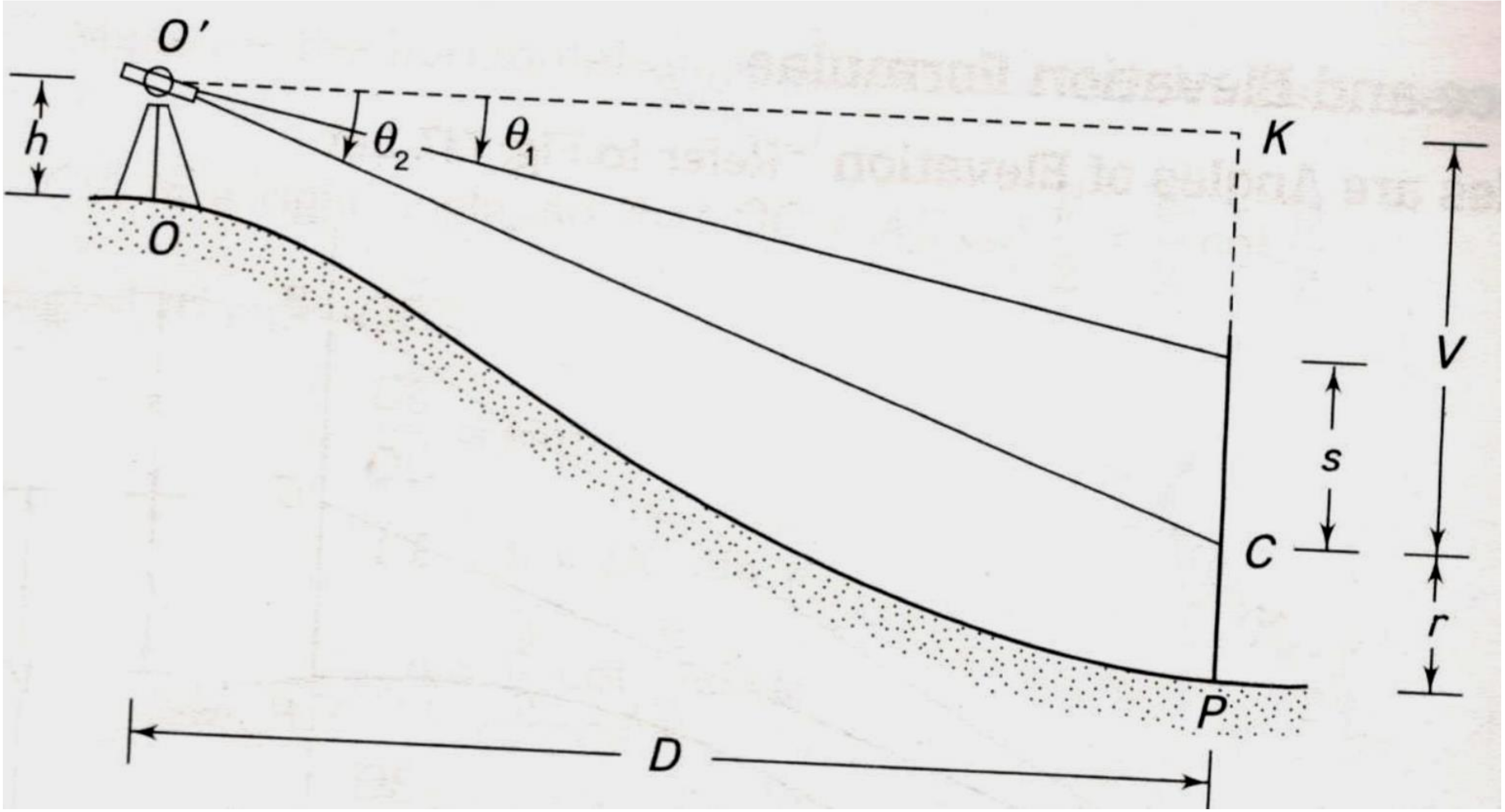
$$D(\tan\theta_1 - \tan\theta_2) = s$$

$$D\ = \frac{s}{\tan\theta_1 - \tan\theta_2}$$

RL of point P = RL of point O +h+ V – r

V- vertical distance from line of collimation to bottom staff reading
r- distance from the foot of the staff to bottom staff reading

Case-II (Both angles are angle of depression)



$$\tan\theta_2 = \frac{V}{D} \Rightarrow V = D\ \tan\theta_2$$

$$\tan\theta_1 = \frac{V - s}{D} \Rightarrow V - s = D\ \tan\theta_1$$

$$D\tan\theta_2 - s = D\ \tan\theta_1$$

$$D\tan\theta_2 - D\ \tan\theta_1 = s$$

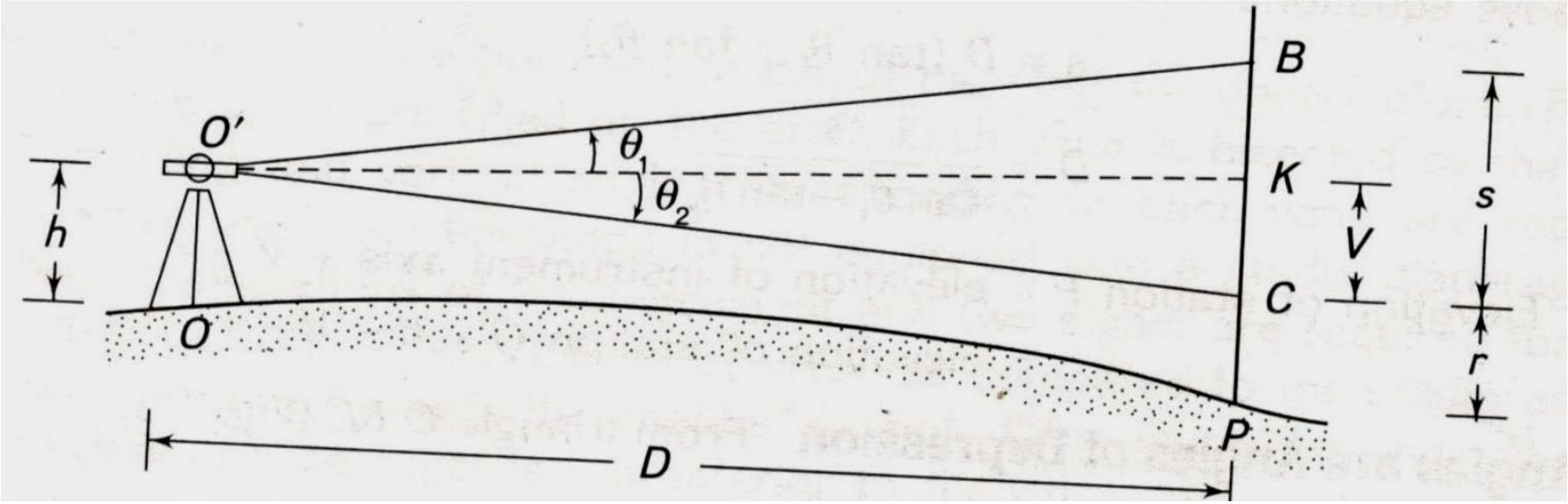
$$D(\tan\theta_2 - \tan\theta_1) = s$$

$$D = \frac{s}{\tan\theta_2 - \tan\theta_1}$$

RL of point P = RL of point O+h - V – r

V- vertical distance from line of collimation to bottom staff reading
r- distance from the foot of the staff to bottom staff reading

Case-III (One angle is angle of elevation other angle is angle of depression)



$$\tan\theta_2 = \frac{V}{D} \Rightarrow V = D\ \tan\theta_2$$

$$\tan\theta_1 = \frac{s-V}{D} \Rightarrow s - V = D\ \tan\theta_1$$

$$s - D\tan\theta_2 = D\ \tan\theta_1$$

$$D\tan\theta_1 + D\ \tan\theta_2 = s$$

$$D(\tan\theta_1 + \tan\theta_2) = s$$

$$D = \frac{s}{\tan\theta_1 + \tan\theta_2}$$

RL of point P = RL of point O + h --V – r

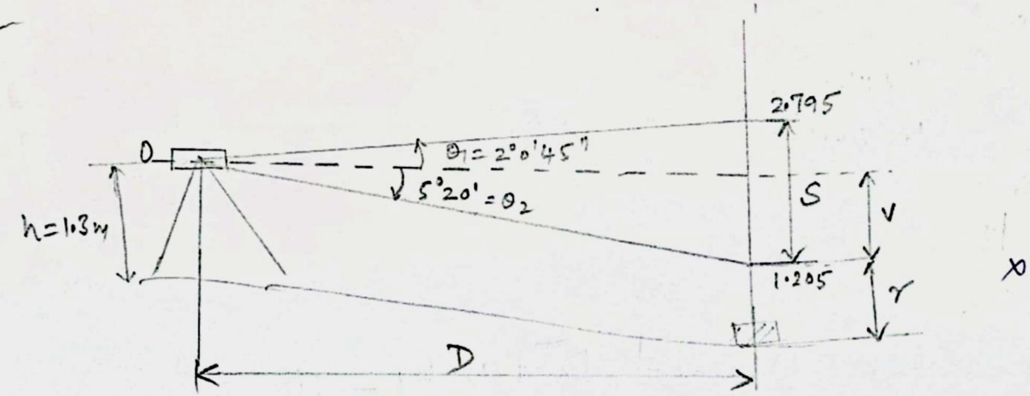
V- vertical distance from line of collimation to bottom staff reading
r- distance from the foot of the staff to bottom staff reading

Q6:A tacheometer is set up at a point O. The observations were taken keeping the stadia rod at points X and Y. The tangential method is followed. The observation table is given below.

Instrument at	Staff at	Vertical angles	Stadia reading in m
O	X	$-5^{\circ}\ 20'$ and $2^{\circ}\ 0'\ 45''$	1.205 and 2.795
	Y	$4^{\circ}\ 22'\ 6''$ and $3^{\circ}\ 32'\ 57''$	3.115 and 1.695

The RL of the instrument station is 120m. The height of instrument is 1.300m. Find the distance OX and OY. Find the RL of X and Y.

(2) Case (i) Instrument at O & Staff at X



$V = D \tan\theta_2$ —(1)

$S - V = D \tan\theta_1$ —(2)

-2- $S = D(\tan\theta_1 + \tan\theta_2)$

$2.795 - 1.205 = D\,(\tan 5°20' + \tan 2°0'45'')$

$D = \underline{12.375\,m}$ (2)

$V = D \tan\theta_2 = 12.375 \tan(5°20')$

-2- $= \underline{1.155\,m}$ (2)

$$RL \text{ of } X = RL \text{ of } BM + h - V - r$$

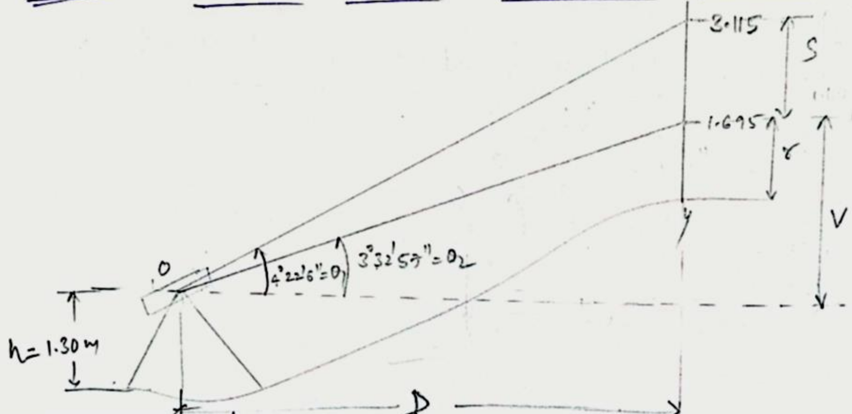
$$= 120 + 1.3 - 1.155 - 1.205$$

1

1

$$= \underline{118.94}$$

Case (2) Instrument at O & Staff at Y



$$~~tan~~ \; (V+S) = D \tan\theta_1 \quad \text{--- (1)}$$

$$V = D \tan\theta_2 \quad \text{--- (2)}$$

(1) − (2)

$$S = D(\tan\theta_1 - \tan\theta_2)$$

$$3.115 - 1.695 = D\left(\tan 4^\circ 22' 6'' - \tan 3^\circ 32' 57''\right)$$

2

$$D = 98.88\,m = OY$$

2

2

$$V = D\tan\theta_2 = 98.88 \tan 3^\circ 32' 57''$$

$$= \underline{6.133\,m}$$

2

1

$$\text{Elevation of } Y = RL \text{ of } BM + h + V - r$$

1

$$= 120 + 1.30 + 6.133 - 1.695 = 125.738\,m$$


Thank You